Taylor Polynomials

MATH 146 – Calculus II

Torin Quinlivan

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Recall: Linear Approximation

Example: Find the linear approximation to $f(x) = \sqrt{x}$ at $a = 4$.

Example: Estimate $\sqrt{3.7}$.

Extending the Linear Approximation

What makes the linear approximation work?

Consider the linear approximation $L(x) = a + bx$ to a function $f(x)$ centered at $x = 0$.

Idea: The linear approximation has the same "zeroth" and "first" derivative as $f(x)$ at the point a.

Question: Suppose $Q(x) = a + bx + cx^2$ is a polynomial of degree two, which has the same zeroth, first and second derivative as $f(x)$ at $x = 0$. What are a, b and c?

Extending the Linear Approximation

Question: What about a cubic polynomial $C(x) = a + bx + cx^2 + dx^3$ and beyond?

Taylor Polynomials

Maclaurin Polynomial

The **Maclaurin polynomial** of degree n for a function $f(x)$ is

$$p_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k.$$

Taylor Polynomial

The **Taylor polynomial** of degree n centered at a for a function $f(x)$ is

$$p_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

Example: Taylor Polynomials

Example: Find the Maclaurin polynomial of degree 4 for $f(x) = e^x$.

Example: Find the Taylor polynomial of degree 2 for $f(x) = \sqrt{x}$ centered at $a = 4$. Estimate $\sqrt{3.7}$.

Example: Taylor Polynomials

Example: Find the Maclaurin polynomial of degree 7 for $f(x) = \sin(x)$.

Example: Find the Taylor polynomial of degree 3 for $f(x) = \cos x$ centered at $a = \dfrac{\pi}{4}$.

Remainder for Taylor Polynomials

Remainder

The **remainder** $R_n(x)$ for the Taylor polynomial $p_n(x)$ for $f(x)$ is the difference $R_n(x) = f(x) - p_n(x)$.

Taylor's Theorem

Taylor's Theorem

Suppose $f(x)$ has all derivatives up to $f^{(n+1)}(x)$ in an interval $\mathcal{I}$ containing a. Then for any x in $\mathcal{I}$, there is a c between a and x such that

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Taylor's Theorem

Consequence of Taylor's Theorem

If $\left|f^{(n+1)}(x)\right| \leq M$ on $\mathcal{I}$, then

$$|R_n(x)| = |f(x) - p_n(x)| \leq \frac{M|x-a|^{n+1}}{(n+1)!}$$

for all x in $\mathcal{I}$.

Example: Error Approximation

Example: Find the maximum error for the Maclaurin polynomial $p_4(x)$ for $f(x) = e^x$ on the interval $[-2, 2]$.

Example: How large will n have to be so that the maximum error for the Maclaurin polynomial $p_n(x)$ for $f(x) = e^x$ on the interval $[-2, 2]$ is less than or equal to 0.00001.

Goal: Taylor Polynomials of Infinite Degree

Question: Can we define $p_\infty(x) = \displaystyle\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k$ so that $p_\infty(x) = f(x)$?

Example: What would $p_\infty(x)$ be for $f(x) = e^x$ at $a = 0$?